Диплом
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Задача:
Условие:
Решение:

$$\hat{H} = g\left(\hat{b}^\dagger_- \hat{a}_+ + \hat{b}_- \hat{a}^\dagger_+\right)$$

$$\left|\psi_{in}\right\rangle = e^{-|\alpha|^2/2} e^{\alpha \hat{b}^\dagger_H} \left|0\right\rangle = e^{-|\alpha|^2/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} \left|n_H\right\rangle$$

$$\left|\psi^H_{in}\right\rangle = e^{\alpha \hat{b}^\dagger_H} \left|0\right\rangle = \sum_n \frac{\alpha^n}{\sqrt{n!}} \left|n_H\right\rangle$$

$$\left|\psi_{out}\right\rangle = e^{-i\hat{H}t} \sum_n \frac{\alpha^n}{\sqrt{n!}} \left|n_H\right\rangle = e^{\alpha \hat{b}^\dagger_+/\sqrt{2}}\ \hat{S}_0\ e^{\alpha \hat{b}^\dagger_-/\sqrt{2}}\ \left|0\right\rangle$$

$$\left|\psi_{out}\right\rangle = \frac{e^{-|\alpha|^2/2}\ e^{\alpha \hat{b}^\dagger_H}}{2} \left[e^{-i\theta/2}\ e^{-i\gamma \hat{b}^\dagger_-} + e^{i\theta/2}\ e^{i\gamma \hat{b}^\dagger_-} + \right.$$
$$\left. + \left(e^{-i\theta/2}\ e^{-i\gamma \hat{b}^\dagger_-} - e^{i\theta/2}\ e^{i\gamma \hat{b}^\dagger_-}\right) \hat{a}^\dagger_+\right] \left|0\right\rangle$$

$$\left|\psi^V_{out}\right\rangle \approx \cos\left(\frac{\theta}{2}\right) \left|0\right\rangle + \frac{i\delta - 1}{\sqrt{2}} \sin\left(\frac{\theta}{2}\right) \left|1\right\rangle + \frac{i\delta}{2} \cos\left(\frac{\theta}{2}\right) \left|2\right\rangle$$

$$\left|\psi_{out}\right\rangle = \exp\left(-i\hat{H}t\right) = e^{\alpha \hat{b}^\dagger_+/\sqrt{2}}\ \hat{S}_0\ e^{\alpha \hat{b}^\dagger_-/\sqrt{2}}\ \left|0\right\rangle$$

$$i\frac{\partial \hat{\rho}_{HV}}{\partial t} = \left[\hat{H}; \hat{\rho}_{HV}\right]$$

$$\hat{\rho}_{HV} = e^{\alpha \hat{b}^\dagger_H} \hat{\rho}_V e^{\alpha^* \hat{b}_H}$$

$$i\frac{\partial \hat{\rho}_V}{\partial t} = \left[\hat{H}_V; \hat{\rho}_V\right]$$

$$\hat{H}_V = \Omega^H_R \left(\hat{d}^\dagger_\uparrow \hat{a}_\uparrow + \hat{d}_\uparrow \hat{a}^\dagger_\uparrow\right) + ig\left(\hat{b}_V \hat{d}^\dagger_\uparrow \hat{a}_\uparrow - \hat{b}^\dagger_V \hat{d}_\uparrow \hat{a}^\dagger_\uparrow\right)$$

$$g << \Omega^H_R$$

$$\left|\psi_{out}\right\rangle = \cos(\Omega_R t)\hat{a}^\dagger_\uparrow \left|0\right\rangle - i\sin(\Omega_R t)\hat{d}^\dagger_\uparrow \left|0\right\rangle + gt\, i\sin(\Omega_R t)\hat{a}^\dagger_\uparrow \hat{b}^\dagger_V \left|0\right\rangle - gt\ \cos(\Omega_R t)\hat{d}^\dagger_\uparrow \hat{b}^\dagger_V \left|0\right\rangle$$

$$\hat{d}^\dagger_\uparrow$$

$$\hat{b}^\dagger_{-t} \hat{a}^\dagger_\uparrow$$

$$\left|\psi_{out}\right\rangle = \cos(\Omega_R t)\hat{a}^\dagger_\uparrow \left|0\right\rangle - i\sin(\Omega_R t)\hat{b}^\dagger_{\uparrow t} \hat{a}^\dagger_\uparrow \left|0\right\rangle + gt\, i\sin(\Omega_R t)\hat{a}^\dagger_\uparrow \hat{b}^\dagger_V \left|0\right\rangle - gt\ \cos(\Omega_R t)\hat{b}^\dagger_{\uparrow t} \hat{a}^\dagger_\uparrow \hat{b}^\dagger_V \left|0\right\rangle$$

$$-i\hat{b}_{Vt}^{\dagger}\hat{a}_{\uparrow}^{\dagger}$$

$$|\psi_{out}\rangle = \cos(\Omega_R t)\hat{a}_{\uparrow}^{\dagger}|0\rangle - \sin(\Omega_R t)\hat{b}_{Vt}^{\dagger}\hat{a}_{\uparrow}^{\dagger}|0\rangle + gt\,i\sin(\Omega_R t)\hat{a}_{\uparrow}^{\dagger}\hat{b}_{V}^{\dagger}|0\rangle - igt\,\cos(\Omega_R t)\hat{b}_{V}^{\dagger}\hat{b}_{Vt}^{\dagger}\hat{a}_{\uparrow}^{\dagger}|0\rangle$$

$$|\psi_{out}\rangle = \cos(\Omega_R t)|0\rangle + (igt-1)\sin(\Omega_R t)|1\rangle + igt\,\cos(\Omega_R t)|2\rangle$$

$$\hat{H}_+ = E_t\hat{d}_{\uparrow}^{\dagger}\hat{d}_{\uparrow} + \Omega_R\left(e^{-i\omega_L t}\hat{d}_{\uparrow}^{\dagger}\hat{a}_{\uparrow} + e^{-i\omega_L t}\hat{d}_{\uparrow}\hat{a}_{\uparrow}^{\dagger}\right) + g\left(\hat{b}_+\hat{d}_{\uparrow}^{\dagger}\hat{a}_{\uparrow} + \hat{b}_+^{\dagger}\hat{d}_{\uparrow}\hat{a}_{\uparrow}^{\dagger}\right) + E_t\hat{b}_+^{\dagger}\hat{b}_+$$

$$\hat{H}_+ = \Delta\hat{d}_{\uparrow}^{\dagger}\hat{d}_{\uparrow} + \Delta\hat{b}_+^{\dagger}\hat{b}_+ + \Omega_R\left(\hat{d}_{\uparrow}^{\dagger}\hat{a}_{\uparrow} + \hat{d}_{\uparrow}\hat{a}_{\uparrow}^{\dagger}\right) + g\left(\hat{b}_+\hat{d}_{\uparrow}^{\dagger}\hat{a}_{\uparrow} + \hat{b}_+^{\dagger}\hat{d}_{\uparrow}\hat{a}_{\uparrow}^{\dagger}\right)$$

$$\hat{H}_- = \Delta\hat{d}_{\downarrow}^{\dagger}\hat{d}_{\downarrow} + \Delta\hat{b}_-^{\dagger}\hat{b}_- + \Omega_R\left(\hat{d}_{\downarrow}^{\dagger}\hat{a}_{\downarrow} + \hat{d}_{\downarrow}\hat{a}_{\downarrow}^{\dagger}\right) + g\left(\hat{b}_-\hat{d}_{\downarrow}^{\dagger}\hat{a}_{\downarrow} + \hat{b}_-^{\dagger}\hat{d}_{\downarrow}\hat{a}_{\downarrow}^{\dagger}\right)$$

$$\hat{H}_B = \alpha B_x\left(\hat{a}_{\downarrow}^{\dagger}\hat{a}_{\uparrow} + \hat{a}_{\uparrow}^{\dagger}\hat{a}_{\downarrow}\right) + \alpha B_z\left(\hat{a}_{\uparrow}^{\dagger}\hat{a}_{\uparrow} - \hat{a}_{\downarrow}^{\dagger}\hat{a}_{\downarrow}\right) + \beta B_z\left(\hat{d}_{\uparrow}^{\dagger}\hat{d}_{\uparrow} - \hat{d}_{\downarrow}^{\dagger}\hat{d}_{\downarrow}\right)$$

$$\hat{a}_{\uparrow}^{\dagger}|0\rangle\ ;\ \hat{a}_{\downarrow}^{\dagger}|0\rangle\ ;\ \hat{d}_{\uparrow}^{\dagger}|0\rangle\ ;\ \hat{d}_{\downarrow}^{\dagger}|0\rangle\ ;\ \hat{b}_+^{\dagger}\hat{a}_{\uparrow}^{\dagger}|0\rangle\ ;\ \hat{b}_-^{\dagger}\hat{a}_{\downarrow}^{\dagger}|0\rangle\ ;\hat{b}_+^{\dagger}\hat{d}_{\uparrow}^{\dagger}|0\rangle\ ;\ \hat{b}_-^{\dagger}\hat{d}_{\downarrow}^{\dagger}|0\rangle$$

$$\hat{b}_-^{\dagger}\hat{a}_{\uparrow}^{\dagger}|0\rangle\ ;\ \hat{b}_+^{\dagger}\hat{a}_{\downarrow}^{\dagger}|0\rangle\ ;\hat{b}_-^{\dagger}\hat{d}_{\uparrow}^{\dagger}|0\rangle\ ;\ \hat{b}_+^{\dagger}\hat{d}_{\downarrow}^{\dagger}|0\rangle$$

$$|\uparrow\rangle + |\downarrow\rangle \to |\uparrow R\rangle + |\downarrow L\rangle$$

$$|\uparrow\rangle + |\downarrow\rangle \to (|\uparrow\rangle + |\downarrow\rangle)|R\rangle + (-|\uparrow\rangle + |\downarrow\rangle)|L\rangle$$

$$|\uparrow\rangle - |\downarrow\rangle \to |+_1+_2\rangle - |-_1-_2\rangle$$

$$\Omega_R^H \sim g\sqrt{N}$$

$$\frac{\partial\hat{\rho}}{\partial t} = -i\left[\hat{H};\hat{\rho}\right] - \hat{L}(\hat{\rho})$$

$$\hat{L}(\hat{\rho}) = \gamma_{PD}\left(\hat{\rho} - \hat{\sigma}_z\hat{\rho}\hat{\sigma}_z\right)$$

$$\hat{\rho}_{in}$$

$$\hat{\rho}_{out} = \hat{S}_{\tau}\left(\hat{\rho}_{in}\right)$$

$$\hat{\rho}_{ph} = \hat{M}_{spin\ relax}\left(\hat{\rho}_{out}\right)$$

$$= f_{+x}\left(B/T\right)\langle x|\hat{\rho}_{out}|x\rangle + f_{-x}\left(B/T\right)\langle -x|\hat{\rho}_{out}|-x\rangle$$

$$f_{+x}\left(B/T\right) = \frac{e^{B/T}}{e^{B/T} + e^{-B/T}}$$

$$f_{-x}\left(B/T\right) = \frac{e^{-B/T}}{e^{B/T} + e^{-B/T}}$$

$$= f_{+x}\left(B/T\right)|x\rangle\langle x| + f_{-x}\left(B/T\right)|-x\rangle\langle -x|$$

$$\sqrt{P}$$

$$|000\rangle + |001\rangle + |010\rangle - |011\rangle + |100\rangle + |101\rangle - |110\rangle + |111\rangle$$

$$F = \left(\langle +_1+_2| + \langle -_1-_2|\right)\hat{\rho}_{out}\left(|+_1+_2\rangle + |-_1-_2\rangle\right)$$

$$\hat{\rho}_{ph} = \hat{\rho}_0 + \hat{\rho}_1 + \hat{\rho}_2$$

$$p_1 = Tr\left(\hat{\rho}_1\right)\ \ ;\ \ p_2 = Tr\left(\hat{\rho}_2\right)$$

$$\hat{\rho}_2^{(2\pi)} \approx (g\tau)^2 \left(\frac{\Omega_R^2}{\Omega_R^2+\Delta^2}\right)^2 \begin{pmatrix} 1 & 0 & 0 & \tanh^2(B/T) \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \tanh^2(B/T) & 0 & 0 & 1 \end{pmatrix}$$

$$g^{(2)}(0) = \frac{2\left|\delta\right|^2 \cos^2(\theta/2)}{\left(\sin^2(\theta/2)+\left|\delta\right|^2\right)^2}$$

$$|0\rangle = |H\rangle$$

$$|1\rangle = |V\rangle$$

$$\phi_{ab}^{+} = \frac{1}{\sqrt{2}}\left(H_a H_b + V_a V_b\right)$$

$$\frac{\mu g B}{T}$$